Scenario-Based Questions | Set 3

1. Launch and test a website locally

Scenario: I want to launch my web development project on my computer and check whether it works properly.

Answer / Logic:

1. Open the project folder in VS Code or a terminal.
2. Install the required packages and check the project settings.
3. Start the local server (for example, python manage.py runserver for Django).
4. Open the local address in a browser, test the pages and forms, and fix any errors.

2. Travel website: destination recommendations

Scenario: A travel website wants to suggest places based on interests such as beaches, hiking and history.

Answer / Logic:

1. Collect users' interests and add relevant tags to each destination.
2. Convert these interests and destination tags into numerical features.
3. Use content-based filtering and cosine similarity to find matching places.
4. Recommend the best matches and improve suggestions using user feedback.

3. Telecom: reduce churn prediction features

Scenario: A telecom company has 150 customer features and wants to predict churn without losing much accuracy.

Answer / Logic:

1. Clean missing values and remove duplicate or constant features.
2. Use feature importance or SelectKBest to identify useful churn predictors.
3. Try RFE with a classification model to reduce the feature count.
4. Compare cross-validation scores before and after selection and keep the smaller reliable set.

4. E-commerce: recently viewed product recommendations

Scenario: An online shopping platform wants to recommend products related to what a customer recently viewed.

Answer / Logic:

1. Track recently viewed products and their category, brand and other details.
2. Convert product information into feature vectors.
3. Use cosine similarity to find products similar to the recently viewed items.
4. Rank recent interests higher and remove products already purchased or unavailable.

5. Fitness tracker: calorie prediction features

Scenario: A fitness app predicts calories burned from heart rate, workout duration, weight and other readings.

Answer / Logic:

1. Check the data for missing values and unrealistic sensor readings.
2. Study how each feature relates to calories burned.
3. Use RFE or Random Forest feature importance with a regression model.
4. Compare prediction errors on validation data and retain the useful features.

6. Marketplace: recommend sellers to new buyers

Scenario: A marketplace wants to suggest sellers to new buyers using their preferences and existing seller ratings.

Answer / Logic:

1. Ask new buyers about product needs, budget and preferred location.
2. Build seller profiles using categories, service details and past ratings.
3. Use content-based matching first, since new buyers have no rating history.
4. Rank suitable sellers using preference match and reliable seller feedback; update with later interactions.

7. Ed-tech: identify useful quiz performance features

Scenario: An ed-tech platform predicts quiz performance from study time, topic difficulty and engagement.

Answer / Logic:

1. Collect study time, quiz results, difficulty and engagement data.
2. Clean the data and remove unrelated or highly repetitive columns.
3. Use SelectKBest or RFE to rank the most useful features.
4. Validate the prediction model and keep the features that improve performance on unseen students.

8. OTT: recommendations from watch completion

Scenario: An OTT platform wants to suggest shows based on how much of each show users finish watching.

Answer / Logic:

1. Record watched shows and completion percentages for each user.
2. Create a user-show interaction table and treat higher completion as a stronger interest signal.
3. Find similar users or shows using collaborative filtering and cosine similarity.
4. Recommend unwatched shows, while considering genre preferences and recent activity.

9. Insurance: simplify claim amount prediction

Scenario: An insurance company uses 120 customer-related features to predict claim amounts.

Answer / Logic:

1. Handle missing values and remove duplicate or low-variance columns.
2. Use correlation checks and regression-based feature importance to shortlist inputs.
3. Apply RFE or PCA to reduce the feature set, depending on whether interpretation is needed.
4. Compare MAE or RMSE on held-out data and choose a simpler model with acceptable accuracy.

10. Food delivery: dietary preference recommendations

Scenario: A food delivery app wants to recommend dishes for vegan, spicy and gluten-free preferences.

Answer / Logic:

1. Store dietary tags and ingredients for every dish.
2. Ask users about food preferences and any dietary restrictions.
3. Filter out dishes that do not meet required restrictions before ranking.
4. Use content-based matching to recommend suitable dishes and learn from user feedback.